Choosing the Number of Principal Components in Optimal-Transport Steering

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TLDR

Nanfack et al. (2026), "Efficient Refusal Ablation in LLM through Optimal Transport", found that PCA-OT gets worse at high K on refusal. I tested whether more training data fixes this, and whether the same K trend holds on a second task, sycophancy. Doubling the refusal training set (128 to 260 prompts) did not clearly help, since the three random 128-prompt draws varied about as much as the doubling did. On sycophancy the trend reversed: PCA-OT had no effect at K=1–10 and only steered at K=20–50. This seems to come from where the signal sits. For refusal, 99.9% of the difference-of-means direction lies in the first PCA direction, while for sycophancy it is spread over many. So the best K depends on the task, and the direction with the most variance may not simply be the direction which splits the two groups.

Intro

Efficient Refusal Ablation in LLM through Optimal Transport (2026) by Nanfack, Belilovsky & Dohmatob creates a new method for steering models, called PCA-Gaussian Optimal Transport (PCA-OT). They found PCA-OT tends to perform better than RFA (Arditi et al., 2024) and AcT (Rodriguez et al., 2025) when the model is steered to stop refusing harmful prompts, as on Llama-2-13B, PCA-OT2 reaches 83.8% attack success rate (ASR) vs 46.5% for RFA and 78.5% for AcT. Steering performance (ASR) of PCA-OT is highest at small K (1-5) values and drops when K=10 (Llama-2-13B: 82.4% at K=2, 76.1% at K=10). It also finds a direct correlation between higher values of K and worse general model performance (higher perplexity)(Pile perplexity 8.5 at K=1, 45.1 at K=10).

Research Question

I decided to test whether the degradation in both steering performance and general model performance at high K would change if the training set size was increased. I also tested if the same trends with high K value also happened on a different steering task (sycophancy).

Test 1: Increasing Training Size

Method

First, I recreated their experiment (on Llama-2-13B (Touvron et al., 2023) and Qwen2.5-14B (Qwen Team, 2024), at K = 1, 2, 5, 10), and did three runs on their training set size, 128, and another run on 260. I did 3 random sample draws from the curated 260-prompt training pool the original paper used for my 128 runs, and for the 260 run, I used the entire 260 curated set.

Results

I got similar results for the 128 runs to their results on Llama-2-13B (Table 1); however, Qwen2.5-14B differed slightly. On the 260 run, I saw no substantial changes. The results of the three independent draws had a spread that was as large as the effect of doubling the data, and the spread grew with K (At K=1-2, Pile perplexity varied by 1-6%, and 22-89% at K=10).

	Paper (n=128)		Mine, n=128 (3 draws)		Mine, n=260	
PCA-OT1	**ASR %**	**Pile PPL**	**ASR %**	**Pile PPL**	**ASR %**	**Pile PPL**
K=1	81.1	8.50	79.2 – 82.4	7.75 – 7.93	83.0	7.82
K=2	82.4	8.59	81.8 – 83.0	7.76 – 7.98	81.8	7.86
K=5	80.5	12.20	81.1 – 83.6	8.35 – 8.56	81.8	8.63
K=10	76.1	45.10	70.4 – 79.9	20.83 – 40.17	74.2	26.91

Table 1. Refusal replication for PCA-OT1 on Llama-2-13B (159 HarmBench prompts (Mazeika et al., 2024), judged by Llama Guard 2 (Meta Llama Team, 2024)). The n=128 columns show the range over three random draws from the 260-prompt pool.

ASR = LlamaGuard attack success rate (higher = stronger steering). Pile PPL = perplexity on Pile text (Gao et al., 2020) (lower = healthier model; unsteered 7.68).

Test 2: Sycophancy Task

Method

The more interesting of the two experiments is applying PCA-OT to a different task. I used the CAA repository (Rimsky et al., 2024), where it included 20,184 A/B question pairs, where one answer is sycophantic, and the other isn't. I ran it on Llama-2-7B-chat, the model CAA itself uses. I had a training set of 1,600 pairs, and 500 test questions and a 300-question validation set, with 3 random training draws.

I chose the layer of each method using the validation set. I then trained and tested the model for 1, 2, 3, 5, 10, 20, 50 values of K, with PCA-OT1 at layer 12, and PCA-OT2 at layers 12-13. I measured how often the model picked the sycophantic answer. As the model was being steered not to be sycophantic, the lower this measure is, the better. I then compared each K ranked vector, which was each PCA direction from 1-50, to the difference of means direction for sycophancy. I did this for both the refusal task, and the sycophancy task.

Results

It was found that PCA-OT1 and 2 performed better as K increased. At K=1-10, the model performed the same as it would when steered by a random direction (both stayed near the unsteered 72%), and it started to move at K=20, until it got K=50 (63% for PCA-OT1 and 52% for PCA-OT2; Figure 1, Table 2). However, PCA-OT2 at K=50 damaged the model, as Pile perplexity nearly doubled (8.86 to 16.7).

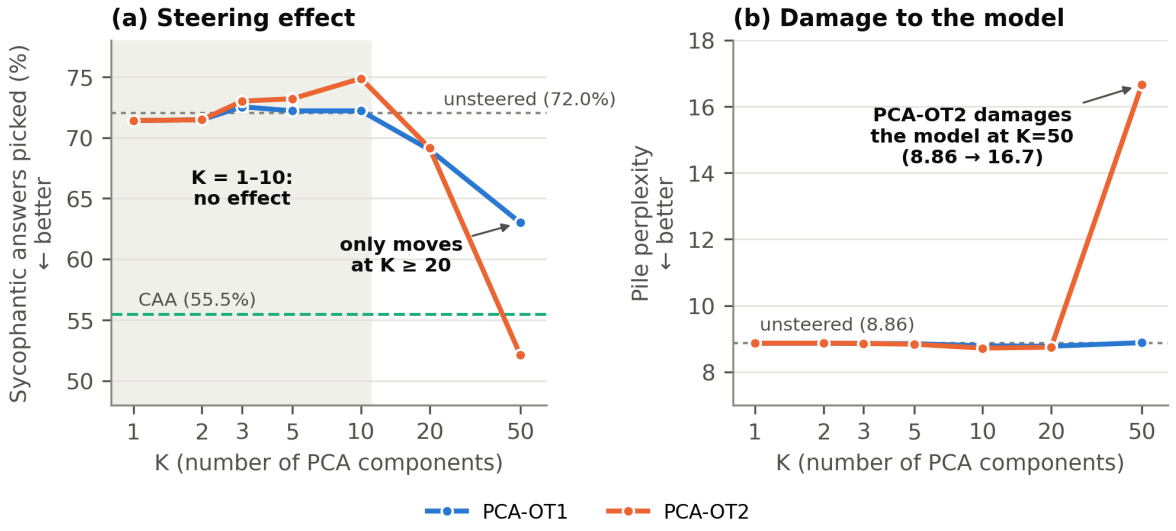


Figure 1. Sycophancy steering on Llama-2-7B-chat versus K (500 test questions, mean of 3 training draws, n=1,600). (a) Share of questions where the model picks the sycophantic answer; lower is better, and the dashed line is CAA. (b) Pile perplexity; lower is better. Dotted lines are the unsteered model.

Method	Sycophantic answers picked	Pile perplexity
Unsteered	72.0%	8.86
Random direction (control)	72.0%	9.00
CAA	55.5%	8.79
RFA	59.2%	8.83
AcT	67.1%	9.04
PCA-OT1 K=2	71.5%	8.86
PCA-OT1 K=50	63.0%	8.88
PCA-OT2 K=2	71.5%	8.87
PCA-OT2 K=50	52.1%	16.66

Table 2. Sycophancy test-set results for each method, Llama-2-7B-chat (n=1,600, mean of 3 draws). Lower is better in both columns. At K=2, PCA-OT is indistinguishable from the random-direction control.

For sycophancy, the direction is spread out over many PCA directions (the first holds 2.7%, and about 20 are needed to capture 93%). For refusal, it has 99.9% of the signal in the first PCA direction (Figure 2).

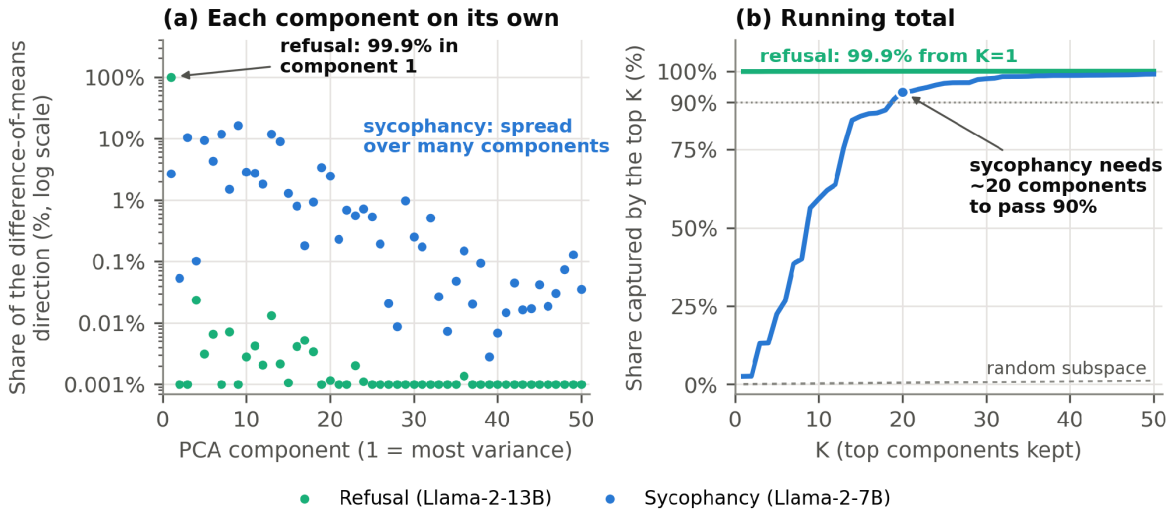


Figure 2. Where the difference-of-means direction sits among the PCA components. (a) Share of the direction along each component on its own (log scale). (b) Share captured by the top K together. The dotted line in (b) is what a random K-dimensional subspace would capture (K/4096).

Discussion

Test 1

This test can neither confirm nor deny if larger samples improve performance of PCA-OT, as for n=260, PPL and ASR fall closely or within the range of the draws from n=128.

Test 2

There is a clear relationship between performance of PCA-OT, and the similarity between PCA K directions and the task's difference of means direction, but it is a threshold rather than a straight line as steering only moved once the top K captured more than about 90% of the direction (Figure 3). However, PCA-OT2 at K=50 also raised Pile perplexity severely (8.86 to 16.7), so its improvement in steering there (52% picking the sycophantic answer, vs 72% unsteered) may just be because the model broke.

It is possible that only certain PCA directions are the ones with the correct signal, and it is not as simple as the highest ranked K is the best. This can be seen, as directions 9, 7, 13, 3, 5 and 14 had the highest similarity to the sycophancy difference of means direction. This should be evaluated further, to see which combination of directions can best optimize for steering and model performance.

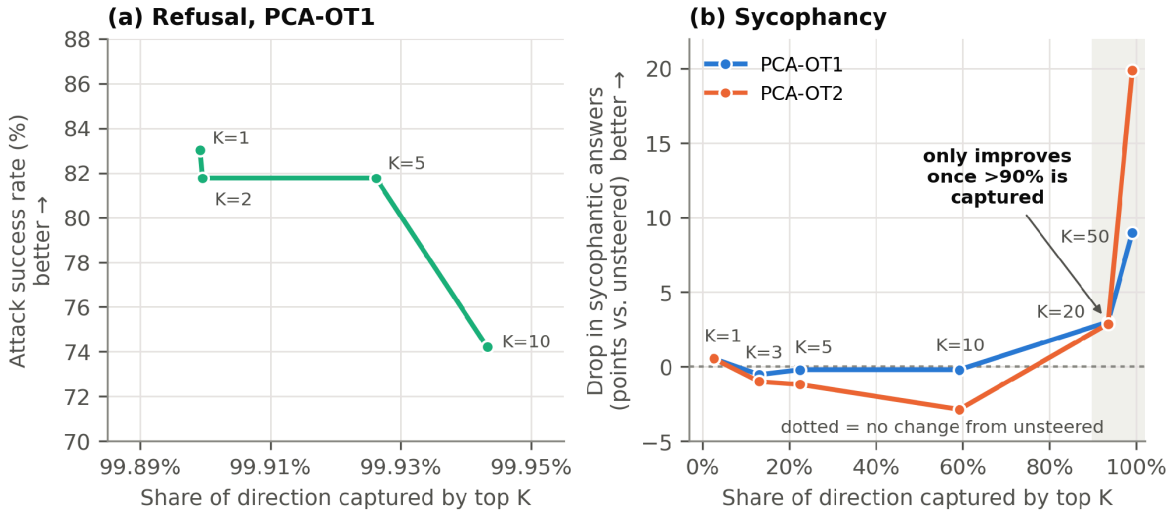


Figure 3. Share of the direction captured by the top K versus steering success; higher is better in both panels. (a) Refusal attack success rate. (b) Drop in sycophantic answers below the unsteered 72%; it only rises once more than 90% of the direction is captured (shaded).

Limitations

The sycophancy results come from one model (Llama-2-7B-chat) and three training draws, and K was only tested up to 50. Test 1 only doubled the training data, so it is inconclusive rather than negative. PCA-OT2 at K=50 nearly doubled Pile perplexity, so its steering gain at that K may reflect a broken model rather than a better direction.

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